
Analysis of Trader Activity vs. Market Sentiment Index

Objective

The objective of this analysis is to examine a dataset of individual trader transactions against a daily market sentiment index (the "Fear & Greed Index"). The goal is to identify patterns in trading activity, such as volume and frequency, based on market sentiment and to explore potential correlations between sentiment, price, and time.

Dataset Summary

Two primary datasets were used for this analysis:

1. **Sentiment Data:** A daily record (2,644 entries) of the Fear & Greed Index. This dataset includes a numeric value (e.g., 10 for "Extreme Fear," 80 for "Extreme Greed") and its corresponding text classification.
2. **Trader Data:** A granular transaction log containing 211,224 trades from 32 unique accounts. This data includes the account, execution price, side (BUY/SELL), and the precise timestamp of each trade.

Data Preparation:

The granular transaction data was merged with the daily sentiment data. This was achieved by converting each trade's timestamp into a simple date, which was then used as a key to join it with the sentiment data for that corresponding day.

The final merged dataset contains 211,218 transactions, with each trade successfully tagged with the market's sentiment classification and value on the day it occurred.

Exploratory Data Analysis (EDA)

(Please paste your three generated charts from the outputs/ folder here.)

Fig 1: Distribution of Trades by Sentiment Classification

Fig 2: Tautological Plot of Sentiment Value vs. Classification

Fig 3: Correlation Heatmap of Numeric Features

Key Insights

Analysis of the merged dataset and visualizations revealed the following insights:

1. Trade Volume is Highest During "Fear"

The "Sentiment Distribution" bar chart (Fig 1) shows that the highest volume of transactions occurred on days classified as "Fear" (61,837 trades). This was followed by "Greed" (50,303 trades) and "Neutral." This suggests that market volatility and uncertainty (fear) may drive more trading activity than periods of extreme greed or extreme fear, which saw lower relative volume.

2. Sentiment Trended Negatively Over Time

The correlation heatmap (Fig 3) reveals a very strong negative correlation (-0.88) between timestamp and value (sentiment). This indicates that over the period observed in this dataset, the general market sentiment has trended downwards (e.g., moving from a state of Greed towards Fear).

3. Note on "Average Value vs. Sentiment" Chart (Fig 2)

The analysis attempted to find a link between trader profit/loss (PnL) and market sentiment. However, the value column (the sentiment score itself, e.g., 80) was mistakenly identified and used as a proxy for a PnL metric. Therefore, the "Average Value vs Sentiment" chart (Fig 2) is tautological: it only plots the average sentiment score for each sentiment category, which simply confirms their definitions (i.e., "Extreme Greed" has a high average value, "Extreme Fear" has a low one). No valid conclusion about trader profitability vs. sentiment can be drawn from this specific chart.

Model Results

A Random Forest Classifier was trained to predict the sentiment classification (e.g., "Fear," "Greed") using the other numeric features in the dataset.

- **Model Accuracy: 100.0%**

Diagnosis (Data Leakage):

This perfect score is a classic sign of data leakage. The feature set provided to the model included the value column, but the target classification is a direct, rule-based derivative of this value (e.g., a value between 25-45 is defined as "Fear").

The model did not find a predictive relationship; it simply learned this hard-coded definitional rule. As a result, this model is not useful for *predicting* future sentiment but serves as a confirmation of the definitional link between the value and classification columns.

Conclusion & Next Steps

This analysis successfully merged transaction data with market sentiment and identified a key behavioral pattern: **trading volume was highest during periods of general "Fear."**

The primary objective of linking sentiment to trader profitability was not achieved due to a flaw in feature selection (value was used instead of closed pnl). To build a meaningful analysis, the following next steps are recommended:

1. **Revise Analysis:** Re-run the analysis using the **closed pnl** column (from the original trader.csv) as the metric for profitability, comparing it against the classification.
2. **Refine Model:** To build a useful predictive model, the value and classification columns must be excluded from the feature set. A better model might try to predict a *future* price movement or a trade side (BUY/SELL) based on *past* sentiment and price action.